Name_____ Date_____ Period_____

LESSON 2-5 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Range and IQR

Maryland Standard 6.SP.B.5c · Learning target – I can find the range and the interquartile range of a data set.

REMEMBER

- **Range** = maximum − minimum. It is the spread of all the data.
- **IQR** (interquartile range) = Q3 − Q1. It is the spread of the middle half.
- The IQR is often less sensitive to extremes than the range. Recompute quartiles when the data change.

WORKED EXAMPLE

Find the range and IQR of 3, 7, 8, 10, 12, 15, 20.

Step 1 Range: 20 − 3 = 17

Step 2 Median 10. Q1 = 7 (of 3, 7, 8). Q3 = 15 (of 12, 15, 20).

Step 3 IQR: 15 − 7 = 8

Answer **Range 17, IQR 8**

WORD BANK

range – biggest number − smallest number
IQR – Q3 − Q1
Q1, Q3 – the middles of the lower and upper halves
minimum – the smallest value

PRACTICE

show your work · label units

1. Find each measure.

a. Range of 14, 6, 21, 9: _____

b. Range of 52, 38, 47, 60: _____

c. IQR when Q1 = 18, Q3 = 31: _____

d. IQR when Q1 = 40, Q3 = 56: _____

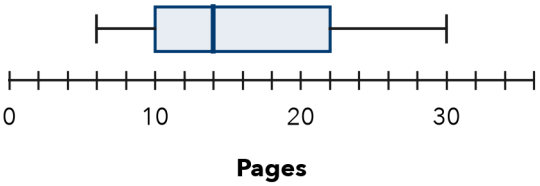
e. Range of 75, 90, 68, 83: _____

f. IQR when Q1 = 7, Q3 = 25: _____

2. **Pages Read.** The box plot shows pages read by students in one week.

Step 1 Read the ends of the whiskers: max − min = range.

Step 2 Read the box edges: Q3 − Q1 = IQR.



a. Range: _____ pages

b. IQR: _____ pages

3. Jump Rope. Seven students jumped rope 35, 52, 41, 60, 28, 47, and 55 times. Find the range and the IQR.

Step 1 Put the 7 numbers in order.

Step 2 Range = last − first.

Step 3 Find Q1 and Q3, then subtract.

a. Range: _____

b. IQR: _____

Show your work.

4. Work Backward. A data set has a **minimum** of 14 and a **range** of 29. Its Q1 is 20 and its IQR is 11.

Step 1 Maximum = minimum + range.

Step 2 Q3 = Q1 + IQR.

a. Maximum: _____

b. Q3: _____

5. Find the Error. Mateo says the IQR is always half of the range. Use the Jump Rope data to explain why he is wrong.

Frame *Half of the range is ___, but the IQR is ___, so Mateo is ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask family members their ages. Find the range of the ages.

Name_____ Date_____ Period_____

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Comparing Variability

Maryland Standard 6.SP.B.5c · Learning target – I can compare the variability of two data sets using range and IQR.

REMEMBER

- More range or IQR = **more variability**.
- Less variability = more **consistent**. Compare range with range, IQR with IQR.
- To describe the middle half when extremes distort the overall spread, consider the IQR and explain your choice.

WORKED EXAMPLE

Team P: IQR 6 points. Team Q: IQR 15 points. Which is more consistent?

Step 1 6 < 15, so Team P's middle half is closer.

Answer **Team P**

WORD BANK

consistent – close together
variability – how spread out the data are
range – max − min
IQR – Q3 − Q1

PRACTICE

show your work · label units

1. Find each IQR.

a. Set P: Q1 = 30, Q3 = 42 _____

b. Set Q: Q1 = 33, Q3 = 39 _____

2. **Basketball.** Points per game: Player 1: 8, 10, 11, 12, 13, 14, 16. Player 2: 2, 6, 9, 12, 15, 18, 22.

Step 1 Range = greatest − least for each player.

Step 2 IQR = Q3 − Q1 for each player.

Step 3 Smaller numbers mean more consistent.

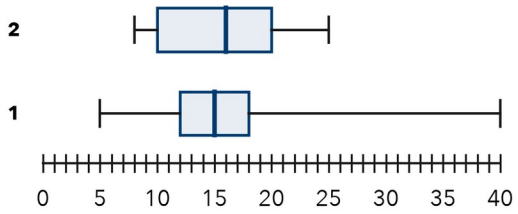
	Range	IQR
Player 1		
Player 2		

More consistent player_____

3. **Spelling Scores.** Group A scored 70, 72, 75, 78, 80. Group B scored 55, 65, 75, 85, 95.

Ranges: A ____ B ____ More consistent group: _____

4. Bus Routes. The box plots show ride times in minutes.



Step 1 Read the whisker ends for each range.

Step 2 Read the box edges for each IQR.

a. Ranges: 1 ____ 2 ____

b. IQRs: 1 ____ 2 ____

5. Jada says Route 2 is more consistent. Kwame says Route 1 is. Who is right? Why?

Frame *___ is right because Route 1's range is big only because ___, and its IQR is ___.*

TEST PRACTICE

choose the best answer

6. Class A's scores have a range of 12. Class B's have a range of 30. Whose scores vary less?

A Class A **B** Class B **C** Both the same **D** Cannot tell

7. What is the IQR of 4, 6, 7, 9, 10, 12, 15?

A 6 **B** 11 **C** 4.5 **D** 9

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___